

Short Questions 2

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Abstract

This paper addresses three foundational concepts in reinforcement learning: the Markov property, value functions, and policy evaluation. The Markov property enables tractable sequential decision-making by ensuring that future states depend only on the present. Value functions $V(s)$ and $Q(s, a)$ quantify expected cumulative rewards for states and state-action pairs, forming the basis for policy optimization. Iterative policy evaluation provides a straightforward algorithm for computing these values through repeated application of the Bellman equation.

Question 1

What does the Markov property of a Markov Process mean?

The Markov property states that the future is independent of the past given the present. The probability of transitioning to the next state depends *only* on the current state, not on the sequence of states that preceded it. Mathematically:

$$P(S_{t+1} | S_t, S_{t-1}, \dots, S_0) = P(S_{t+1} | S_t)$$

This “memoryless” property means the current state contains all information needed to predict future states—the history provides no additional predictive value (Sutton and Barto 2018). As Sutton and Barto note, a state signal that succeeds in retaining all relevant information is said to be *Markov*, or to have the Markov property.

Question 2

What are $V(s)$ and $Q(s, a)$ values in the context of a Markov Decision Process?

$V(s)$ is the **state-value function**: the expected cumulative discounted reward when starting in state s and following policy π thereafter (Sutton and Barto 2018):

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s \right]$$

$Q(s, a)$ is the **action-value function**: the expected cumulative discounted reward when starting in state s , taking

action a , and then following policy π :

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s, A_0 = a \right]$$

The relationship between them is: $V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s, a)$.

Question 3

Briefly describe a naïve algorithm for computing $V(s) \forall s \in S$ in a Markov Decision Process.

First-Visit Monte Carlo estimates $V(s)$ by averaging the returns observed after visiting each state across many episodes (Sutton and Barto 2018):

- 1: Initialize $V(s) = 0$ and $\text{visits}(s) = 0$ for all $s \in S$
- 2: **for** $i = 1$ to num_episodes **do**
- 3: Generate episode using policy π :
 $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T$
- 4: **for** each state S_t in the episode (first visit only) **do**
- 5: $G_t \leftarrow R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^{T-t-1} R_T$
- 6: $V(S_t) \leftarrow \frac{\text{visits}(S_t) \times V(S_t) + G_t}{\text{visits}(S_t) + 1}$
- 7: $\text{visits}(S_t) \leftarrow \text{visits}(S_t) + 1$
- 8: **end for**
- 9: **end for**
- 10: **return** V

This algorithm is considered “naïve” because it can be very slow—in large environments, many episodes are needed to visit all states sufficiently. It may also fail to converge if exploration is inadequate, as some states may never be visited under a deterministic policy. More efficient methods include Temporal Difference learning (which updates after each step rather than waiting for episode completion) and function approximation (which generalizes to unvisited states).

AI Use Statement

Claude (Anthropic) was used to format my answers in AAAI LaTeX format.

References

Sutton, R. S.; and Barto, A. G. 2018. *Reinforcement Learning: An Introduction*. Cambridge, MA: MIT Press, 2nd edition. Available free at <http://incompleteideas.net/book/the-book-2nd.html>.